SUPPLEMENT: EXECUTION-PATH QUALIFICATION AND REALIZED COSTS IN SPECULATIVE DECODING ON CONSUMER SYSTEMS

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A SUPPLEMENT SCOPE

This separate appendix accompanies the measured hardware, mechanism, and runtime studies. It supplies statistical, measurement-boundary, and provenance details. Identifying repository metadata and internal source paths remain outside the anonymous PDF; the nonidentifying B570 prediction-file content digest appears below.

B FROZEN ESTIMATORS

Table A1 preserves each campaign's estimator. The primary sample unit is the prompt. Repeated observations contribute to its summary, rather than being treated as independent prompts. Pair ratios divide matched baseline decode time by speculative decode time. Historical M4, RTX, and B570 counts are text-path agreement from parsed CLI output: M4 HumanEval includes the prompt in its parsed text; B570 strips and normalizes the prompt before re-tokenization. Neither is direct generated-ID equality. The 358H runner and prospective M4 study record generated IDs directly. A geometric mean across prompt summaries is used for headline speedup. B570 timing remains diagnostic regardless of its estimator.

The B570 finalizer builds its prompt vector in the insertion order of the archived pair records. Sorting by task ID changes the finite fixed-seed bootstrap endpoints even though it leaves the underlying estimator unchanged. Exact replay retains that original order. The M4 HumanEval lower endpoint uses a different order statistic from M4 24Q and RTX; those frozen conventions are retained. None of these prompt intervals captures the full distribution of possible thermal, session or runtime states.

C ENERGY BOUNDARIES AND ALLOWED CLAIMS

Ratios above one favor speculation within the named boundary. For M4 fixed-$k = 2$, the package ratio is 1.013 [0.990, 1.034], so a package-energy improvement is not established. The registered $p_{\min} = 0.9$ secondary condition has a package-efficiency ratio 1.041 [1.018, 1.065]. This is not a replacement for the primary condition and does not establish a material additional speed gain. B570 serialized board efficiency is 0.302 [0.285, 0.320]; correctness failure restricts it to diagnosis. No between-platform energy ranking is valid.

D ROUND ACCOUNTING AND STATE SEMANTICS

The 358H target keeps its low-level context across rounds. Candidate evaluation, invalid-suffix rollback when needed, and correction or bonus decoding belong to the complete target-side timing region. Every OpenVINO proposal call receives the entire committed prefix. Its synchronous timing combines reconstruction and proposal; neither term can be isolated from this experiment without changing the measured execution.

The CPU path records 11,520 committed output tokens over 4,677 rounds; the NPU path records 11,520 over 4,686 rounds. Their directly measured C values are 2.4631 and 2.4584. Using the arithmetic baseline time per output token and arithmetic component totals yields CPU $(D, R, X) = (0.4304, 2.9767, 0.00124)$ and NPU $(4.1251, 2.9435, 0.00130)$. The resulting ratios 0.7227 and 0.3477 match ratios of total times, not the prompt-GM headline estimator.

Historical M4/RTX accepted-draft counters exclude correction or bonus output and have output-boundary effects. They are not substituted for the directly recorded

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Table A1. Frozen statistical policies. GM = geometric mean. Indices below are zero-based order statistics of sorted bootstrap draws; NumPy quantiles use the original analyzer's convention.

Campaign	Within prompt	Resamples	Seed	CI endpoints / generator
M4 24Q	GM of 3 ratios	10,000	20260907	[249,9749]; Python Random
M4 HumanEval	GM of 3 ratios	100,000	20260910	[2500,97499]; Python Random
RTX HumanEval	GM of 3 ratios	10,000	20260910	[249,9749]; Python Random
B570 fallback	GM of 3 ratios	10,000	20260910	[250,9750]; Python Random
358H Phase 3	Median of 3 ratios	10,000	20260919	Quantiles .025/.975; NumPy default_rng

Table A2. Sensor boundaries and allowed interpretations. No cells are combined into whole-system energy.

Campaign	Measurement boundary	Allowed statement
M4 HumanEval	Silemetry CPU + GPU + ANE package aggregate; separate DRAM excluded	Fixed-$k = 2$ package-energy interval includes parity; registered threshold variant secondary
RTX 5070 Ti Laptop	NVIDIA board / CPU package are distinct domains; follow-up process window includes loading	No formal serialized HumanEval energy comparison
B570 fallback	Arc adapter board only; no CPU package or wall power	Diagnostic board efficiency only; no workload-global text-path agreement
358H Phase 3	No qualified end-to-end energy boundary	No end-to-end CPU claim; NPU energy not admissible

358H committed-progress metric. Likewise, target-forward PP2/PP1 ratios are never presented as complete realized target-round cost.

E Deviations and Provenance Boundaries

The M4 corrected synthetic ceiling was preregistered but not executed. Earlier synthetic acceptance diagnostics are not a substitute. The representative and separately preregistered code workloads stay distinct. Early MLX precision/correctness failures are qualification history, not failures of the final Metal Q8_0 path.

The B570 formal v2 correctness failure triggered its stop rule. A later six-prompt serialized probe passed 18/18 text-path comparisons, but full-workload serialization yielded only 60/90 under the same comparator. Earlier remote-interaction-contaminated verification observations are excluded from the plotted final serialized cells. A legacy "exact" field label in historical artifacts does not establish workload-global equality.

Windows follow-up derived tables and numerical captures support the manuscript, but historical raw per-run provenance is weaker. The CUDA score capture came from a separate same-commit instrumented build and does not replace formal measurements. The 358H source reconstruction remains `STRONGLY_SUPPORTED`, not proven exact historical upstream identity. Its completed Phase-3 records supersede the blocked-preflight empty scaffold. These are retained qualifications, not unresolved invitations to alter frozen experiments.

The historical RTX HumanEval/106 CUDA reproducer selects token 2790 under sequential/serialized execution and 2629 under batching in three of three runs at the same location. A separate same-commit diagnostic build captures the score-margin change from $+0.080948$ to -0.126457. This supports stable batch-shape-dependent greedy argmax divergence in that historical configuration; it does not establish a specific kernel, universal CUDA defect, or a shared RTX/B570 root cause.

F Verification Context Lookup

Table A3 retains all historical context cells behind Panel A of the main verification figure. Values are target-forward ratios, not complete target-side round cost or integrated speedup. Context ranges do not represent uncertainty intervals. RTX denotes RTX 5070 Ti Laptop throughout this appendix.

Table A3. Historical target-forward lookup. Each row is a measured context cell; ratios are shown to six decimal places from the frozen source. Status refers to historical text-path agreement, not fresh direct-ID qualification. M4 and B570 contexts are prefix lengths; RTX uses reported context depth. The stacks and contexts are not matched. Rounded ranges are M4 Metal 1.02–1.08, RTX batched 0.98–1.01, RTX serialized 2.14–2.43, B570 fast 1.24–1.64, and B570 serialized 1.98–1.99; they span contexts, not confidence intervals.

Path	Context (tokens)	V_2^{micro}	Historical text-path status
M4 Metal	512	1.022182	Text-path agreement; shared path
M4 Metal	2048	1.076126	Text-path agreement; shared path
M4 Metal	8192	1.032264	Text-path agreement; shared path
RTX batched	0	1.009030	Text-path blocked; diagnostic
RTX batched	256	0.983148	Text-path blocked; diagnostic
RTX serialized	0	2.430996	Text-path secondary control
RTX serialized	256	2.137011	Text-path secondary control
B570 fast	512	1.242576	45/90 text-path; diagnostic, inadmissible
B570 fast	2048	1.637295	45/90 text-path; diagnostic, inadmissible
B570 serialized	512	1.982668	60/90 text-path; diagnostic, inadmissible
B570 serialized	2048	1.991707	60/90 text-path; diagnostic, inadmissible

G Evidence Inventory

Table A4 collects the observed regimes and evidence classes. These are case descriptions, not a hardware ranking, population-wide failure rate, or exhaustive taxonomy. M4's separate-session studies establish opposite observed outcomes; neither diagnosed mechanism is assigned to that contrast.

Table A4. Evidence inventory with row-level output endpoints. P: primary; S: secondary; M: mechanism; D: diagnostic. Historical M4/RTX/B570 rows report text-path agreement; 358H reports direct-ID sequence equality. Both B570 paths remain inadmissible diagnostics. The RTX serialized controls each have 90/90 historical text-path agreement. The 358H P/M entries separate primary outcomes from mechanism-supported decomposition. None establishes functional code correctness; annotations are descriptive and non-exhaustive.

Case	Evidence endpoint / gate	Outcome	Class	Case annotation
M4 representative 24Q	Text-path 72/72	Negative	P	Workload-conditioned
M4 HumanEval30	Text-path 90/90	Profitable	P	Workload-conditioned
RTX batched	Text-path; BLOCKED	Not admissible	M	Verification-constrained
RTX serialized controls	Text-path 90/90 each	Negative	S	Verification-constrained
B570 fast	Text-path 45/90	Not admissible	D	Correctness-blocked
B570 serialized	Text-path 60/90	Not admissible	D	Correctness-blocked
358H CPU	Direct-ID 90/90	Negative	P/M	Target-side dominated 87.34% target side
358H NPU	Direct-ID 90/90	Negative	P/M	State-interface bound 58.35% replay + proposal

H RUNTIME STUDY REPRODUCIBILITY

The M4 runtime estimator first takes each condition's median of 3 repetitions, then forms a prompt latency ratio and its geometric mean across 60 prompts. The offline oracle uses the smaller target-only or Always-Spec median for each prompt. Bootstrap intervals use 10,000 draws with Python Random seed 20260924, resampling prompts in the frozen held-out order. Percentile endpoints linearly interpolate the sorted bootstrap draws at quantiles .025 and .975. A single random stream computes Target/Guard, Target/Always-Spec, and Guard/oracle intervals in that order. This preserves the frozen finite-bootstrap calculation; it does not replace the hardware estimators in Table A1. Intervals condition on the selected policy and observed prompt medians; they do not incorporate recalibration or between-session variation. The oracle selects between two measured fixed conditions, not all possible switching policies, so it is not a universal latency lower bound.

Calibration, qualification, and held-out IDs are disjoint. Selection ranks tasks with no recorded prior M4 exposure in the frozen inventory by a seeded hash within each of HumanEval and Spec-Bench; each split is balanced across these two families. The registered grid has nine candidates: first decision round $E \in \{2, 4, 8\}$ and margin $m \in \{0, 0.05, 0.10\}$. Calibration selects the exact candidate with the lowest geometric-mean latency ratio to the offline oracle. Exact ties break by mean measured guard-compute overhead, smaller exploration count, larger margin, and lexicographic candidate ID. The frozen winner is `E8_M000`; held-out results never select or retune it.

The portable M4 archive contains original observations with generated-token IDs, complete-attempt manifests, qualification and calibration records, frozen source/build information, and analysis. An independent offline verifier reproduces policy selection, qualification and formal direct-ID equality, prompt medians, ratios, intervals, and fallback fraction without executing the models. No runtime energy result is claimed.

The reviewer-requested Guard/Always-Spec contrast is post-study paired secondary analysis, not the registered primary endpoint. Aggregate results were already known; the committed analysis plan is a procedural timestamp, not preregistration or blinding. For each frozen held-out prompt i, form $r_i = \mathrm{median}(T_{\mathrm{Guard},i}) / \mathrm{median}(T_{\mathrm{Always},i})$ over its 3 repetitions, then compute $\exp(60^{-1} \sum_i \log r_i)$. Values above one mean Guard is slower. A fresh standalone Python Random stream with seed 20260924 resamples 60 paired prompts per draw in frozen held-out order, using 10,000 draws and the same linear percentile interpolation; it does not continue the primary stream. The ratio is 1.0271 [1.0130, 1.0436], with the interval above parity. Descriptive prompt counts are 24 Guard-faster, 36 Always-Spec-faster, and 0 exact ties. Both new implementations reconstruct raw medians from the same hash-verified archive and match the primary medians exactly; an independent calculation reproduces both finite-bootstrap endpoints. No new measurement or analysis choice after seeing the result was used. The interval excludes session and recalibration variation; source hashes and all prompt ratios remain in the private evidence ledger.

A metadata-recording defect requires explicit provenance recovery. Loop-variable shadowing caused the frozen helper to return the build-manifest path rather than the calibration-freeze path. Qualification and formal-start records therefore contain the build-manifest digest in the intended freeze-hash field. A versioned recovery analyzer verified the actual freeze against the clean formal-start commit, qualification, frozen source/model/calibration/policy hashes, and complete attempts before applying the original formulas. The archive retains the original markers and pipeline STOPPED state. This is a metadata-recording defect, not a measurement mutation.

The 358H diagnostic is pinned separately from the hardware campaign. Its compact evidence records reproducible target-only same-prefix divergence and proposal playback without live NPU inference, but does not establish a specific rollback-boundary bug. It explicitly records that no formal runtime performance data exist and that failed qualification prevented formal runtime performance evaluation. Internal evidence records retain the exact source commit and file hashes; the local raw diagnostic traces are not included in that compact record. No 358H runtime speedup is inferred from these diagnostics.

I PROSPECTIVE DIRECT-ID QUALIFICATION

Table A5 reports fresh direct-ID qualification separately from the historical text-path lookup and timing inventory. Both studies freeze the same HumanEval30 IDs, Qwen3-8B and Qwen3-0.6B Q8_0 models, greedy $k = 2$, three repetitions, cap 128, and a 90/90 complete direct-ID sequence-equality rule. Prompt IDs and uncommitted proposals are excluded; an earlier generated terminal EOG is included. No mismatch is discarded and no subset qualifies. Historical path-control outcomes informed the replications; prospective freezing does not imply blinded discovery. Each study completes 270 executions; independent offline implementations reproduce all 180 comparisons, including first divergences. Operational elapsed fields do not establish formal performance.

The RTX original embedded-server CLI exposes native committed-token debug events without a source patch. It uses context 4096, reference target micro-batch one, candi-

Table A5. Fresh direct-ID qualification studies. Each path must achieve 90/90 complete generated-ID sequence equality; a wholly equal prompt matches in all three repetitions. QUALIFIED/BLOCKED is the frozen gate result. These studies have no matching fresh latency endpoint.

Platform	Path	Direct-ID pairs	Equal prompts	Gate	Evidence endpoint
RTX 5070 Ti Laptop	Batched	75/90	25/30	BLOCKED	Direct-ID sequences
RTX 5070 Ti Laptop	Serialized	90/90	30/30	QUALIFIED	Direct-ID sequences
B570	Fast	60/90	20/30	BLOCKED	Direct-ID sequences
B570	Serialized	78/90	26/30	BLOCKED	Direct-ID sequences

date micro-batches two/one, driver 616.92, and full CUDA0 offload. The historical RTX driver was 616.56; this is not an exact replay of historical host state. The incomplete RTX v1 transport attempt is preserved and supplies no v2 formal pair; completed v2 uses lossless byte transport, a new prospective freeze, and a complete fresh schedule.

B570 uses source instrumentation to record committed IDs in sidecars, context 1024, and a separately frozen completion reference. Its endpoint selects the first 128 committed IDs or the inclusive earlier EOG; raw CLI overshoot is retained. Vulkan0 uses driver 32.0.101.9033 with full target 37/37 and draft 29/29 layer offload. The initial Task Scheduler launch failed during CP936/UTF-8 environment-metadata decoding before inference, with zero formal executions. The first data-producing attempt directly ran the unchanged frozen runner: no semantic, code, binary, model, driver, or parameter change occurred. This deployment deviation is preserved in provenance.

The runtime/front-end, reference, and context differences preclude a causal hardware comparison. Neither study estimates population failure rates or functional code correctness. Repeat stability holds for every prompt in each condition, but stable disagreement remains disagreement. Historical B570 fast 45/90 and serialized 60/90 text-path counts and historical RTX controls retain their original comparators. New qualification does not retrospectively authorize old timing as strict-ID performance. Portable validators hash included evidence and check recorded identities of omitted external weights/binaries; they do not rehash omitted assets locally.

J B570 TARGET-PARTITION PREDICTION

The B570 Vulkan follow-up uses Qwen3-8B/0.6B Q8_0, greedy $k = 2$, context 1024 and 128 committed IDs or earlier inclusive EOG. Its target-only harness matches the UB128 reference's input IDs, sampler and APIs; H1 changes only target n_{ubatch} to one. The harness reproduces 90/90 reference sequences before comparison with live serialized speculation.

Selection and freeze. The audit covers 8,706 retained file records and archive members; any B570 association is conservatively excluded, even if execution is uncertain. This excludes 31 of 164 tasks. The 24 reserved IDs have the smallest SHA-256 digest of a fixed study prefix plus each eligible task ID; outcomes play no role. Discovery uses exposed HumanEval/89, 101, 104, 162, 3, 10, 68, 70, 19, 20, 28, 32. The frozen tree selects H1 at 36/36; only failure permits H2 fixed-proposal replay (also 36/36), and failure of both stops validation. H1 passes; H2 is not built. Predictor, endpoint, reserved IDs, binaries and criterion are committed before reserved inference.

Prospective validation. The 24 reserved IDs are HumanEval/58, 156, 134, 42, 122, 114, 93, 115, 151, 63, 125, 133, 43, 139, 77, 35, 27, 34, 145, 82, 154, 78, 96, and 138. Target reference and target-only predictor run three times each (144 executions) with within-condition repeat stability. All 72 predicted complete ID arrays and their digest are committed before the first of 72 live serialized outcomes. Primary mechanism validation requires 72/72 predictor-to-live complete-ID agreement; no partial threshold or confidence interval is substituted. That agreement is 72/72. Twenty prompts are predicted reference-equal; four are predicted divergent, and all 24 classifications match. Among the 12 divergent repetition pairs, the first-divergence index matches 12/12. Those four prompts and zero-based indices (reference token $\rightarrow$ serialized token) are HumanEval/134: 88 (914$\rightarrow$1946); /42: 73 (432$\rightarrow$1817); /145: 103 (1184$\rightarrow$686); and /138: 93 (14374$\rightarrow$9064). Reference-to-live direct-ID qualification is 60/72 and BLOCKED on this reserved set.

An independent validator passes 5,681 input, output, hash, endpoint and chronology checks; extracted archive validation passes. No prediction edits, selective reruns, parameter changes, performance or energy experiments occur after the relevant freezes. No-known-exposure is limited to audited records. The result establishes tested partition sufficiency, not a unique kernel, universal rule, CUDA/B570 shared cause or production certificate.

Reviewer-visible commitment record. The comparison uses complete generated-token-ID arrays; prompt IDs are ex-

Table A6. Reference trajectory agreement versus frozen UB1 predictor agreement on the same 24 reserved prompts × 3 repetitions. Predictor/live complete-sequence equality is the primary mechanism-validation endpoint; reference/live equality is an already-observed comparison. The 60/72 reference result remains BLOCKED under its qualification contract.

Comparator to live serialized output	Complete equality
Frozen UB128/reference trajectory	60/72
Frozen UB1 target-only predictor	72/72

cluded. At most 128 committed generated IDs are compared, including an earlier terminal end-of-generation (EOG) ID when it ends generation. No mismatch is discarded. Equality requires the complete sequences, including lengths, to match; first-divergence indices are zero-based.

Content identity. The 72 frozen complete prediction arrays are bound to the following prediction-file content digest (SHA-256; two consecutive 32-character lines):

```
0201ccd1f67492510d34be567aa83f1c
38d3bd1f162a970c05d3cea9d3957826
```

The digest identifies the frozen contents; the temporal ordering below comes from the recorded study chronology rather than from the digest alone.

Recorded event	Time relative to H1 freeze
H1 freeze	T_0
72 prediction arrays finalized	$T_0 + 1722.38\,\mathrm{s}$
First live reserved outcome	$T_0 + 1813.75\,\mathrm{s}$

The recorded study chronology fixes the prediction file before the first live serialized reserved outcome, with a 91.37 s interval between prediction completion and live start. Identifying repository metadata are omitted for double-blind review.

Informative reference-divergent prompts. These cases are reference-divergent but predictor/live-identical.

Task	First index	Ref → Pred/Live ID	Pred=Live
HE/134	88	914 → 1946	3/3
HE/42	73	432 → 1817	3/3
HE/145	103	1184 → 686	3/3
HE/138	93	14374 → 9064	3/3

Across these 4 prompts, the predicted and observed first-divergence positions agree in 12/12 repetition pairs.

The separate anonymous sequence supplement contains complete reference, predictor, and live generated-ID arrays for the 72 reserved runs, with a deterministic equality and divergence checker. It contains no identifying machine or repository metadata. Sanitizing the export changes its bytes, so it cannot reproduce the original prediction-file digest or independently prove the reported temporal freeze; the PDF's digest and chronology have those distinct stated roles.

Table A7. B570 reserved commitment ledger. All 24 prompts have three repeat-stable 128-ID outputs. Pred=Live and Ref=Live count complete generated-ID equality across three runs. Each displayed digest is a 12-hex prefix of the full SHA-256 shared by predictor and live output; complete arrays and hashes are in the anonymous sequence supplement. The first reference-to-live divergence index is zero-based; – means reference equality.

Task	Live len	Pred=Live	Ref=Live	Pred/Live digest	Ref → Live first divergence
HE/58	128	3/3	3/3	`cc12b5e1b600`	–
HE/156	128	3/3	3/3	`b8060d33b72c`	–
HE/134	128	3/3	0/3	`fc12c8453de5`	88: 914 → 1946
HE/42	128	3/3	0/3	`c737973a39a6`	73: 432 → 1817
HE/122	128	3/3	3/3	`ed3172303704`	–
HE/114	128	3/3	3/3	`16037f527b2f`	–
HE/93	128	3/3	3/3	`4b69a84d3675`	–
HE/115	128	3/3	3/3	`43fbb9bd3da9`	–
HE/151	128	3/3	3/3	`f371e4f2b94e`	–
HE/63	128	3/3	3/3	`b3141572ec95`	–
HE/125	128	3/3	3/3	`689af3f02f2d`	–
HE/133	128	3/3	3/3	`c1ed3bf83bde`	–
HE/43	128	3/3	3/3	`2e3776c7ff02`	–
HE/139	128	3/3	3/3	`23a77663974d`	–
HE/77	128	3/3	3/3	`dea569b38784`	–
HE/35	128	3/3	3/3	`93498855fe64`	–
HE/27	128	3/3	3/3	`f0f161a1b9e3`	–
HE/34	128	3/3	3/3	`813cb3180ab5`	–
HE/145	128	3/3	0/3	`31246ffe07b9`	103: 1184 → 686
HE/82	128	3/3	3/3	`a356c9235624`	–
HE/154	128	3/3	3/3	`0743671d064c`	–
HE/78	128	3/3	3/3	`a685f47dbab9`	–
HE/96	128	3/3	3/3	`3824c19fc287`	–
HE/138	128	3/3	0/3	`9e54992a7160`	93: 14374 → 9064